



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

ESSENTIALS OF QUANTUM COMPUTING [CSE-40B1]

Marks: 50

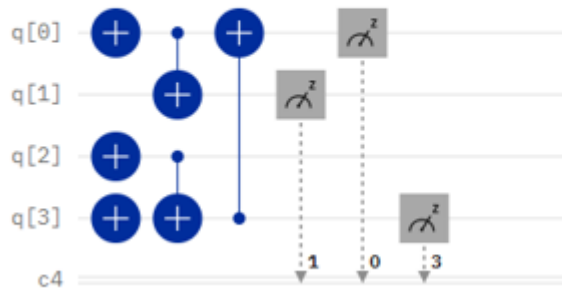
Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A quantum system is in the state, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with α real and β complex. If the probability of measuring $|1\rangle$ is 0.36.
- A) Apply quantum state normalization and probability to:
1. Calculate the magnitude $|\beta|$ based on the measurement probability. (5)
 2. Determine the possible values of α , given that the state must be normalized.
 3. Justify each step based on relevant quantum principles.
- B) Analyze the role of the quantum data plane in a quantum computer. What are its key components and how does it interact with qubits? (5)
- 2) Consider the given postulates:
- Postulate 1: Quantum states are vectors in Hilbert space.
- Postulate 2: Time evolution is governed by a unitary operator acting on these vectors.
- A) Analyze how the linear algebraic structure of Hilbert space supports the first and second postulates of quantum mechanics. Explain why is this structure crucial? (5)
- B) Analyze whether the Pauli-X and Hadamard gates commute (Hint: $XH=HX$) by examining their sequential action on the two-qubit basis states $|00\rangle$ and $|11\rangle$. Determine how changing the order of gate operations affects the final state and interpret the results in terms of phase and interference behavior in quantum circuits. (5)
- 3) **Determine whether the matrix U is unitary**
- A)
$$U = \frac{1}{2} \begin{bmatrix} 1 & 1 & \sqrt{2} \\ 1 & -1 & 0 \\ \sqrt{2} & 0 & -1 \end{bmatrix}$$
 (5)
- B) (5)

Analyze the given quantum circuit and find quantum state at each level.



- 4) Analyze how the matrix representation of a quantum operator changes with the choice of basis.
- A) A linear operator A acts on a 2D quantum system such that:
 $A|0\rangle=|1\rangle$, $A|1\rangle=|0\rangle$.
1. Determine the matrix representation of A in the computational basis $\{|0\rangle,|1\rangle\}$. (5)
 2. Re-express the operator in the Hadamard basis $\{|+\rangle,|-\rangle\}$, Analyze and interpret the change in representation, and discuss what it reveals about the operator's action and eigen structure.
- B) Explain Quantum Entanglement? Analyze how specific combinations of quantum gates-such as Hadamard, CNOT, and CZ-can be used to create entanglement between qubits. Using the example of Bell state generation, explain why certain gate sequences result in entangled states while others do not. (5)
- 5) Given a qubit initialized in the $|+\rangle$, $|-\rangle$ state, analyze the effect of applying an $R_x(\pi)$ followed by $R_y(\pi)$ gate. What is the final state and what is its geometric interpretation? (5)
- A)
- B) Explain why topological qubits offer a promising route toward fault-tolerant quantum computing. (5)

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